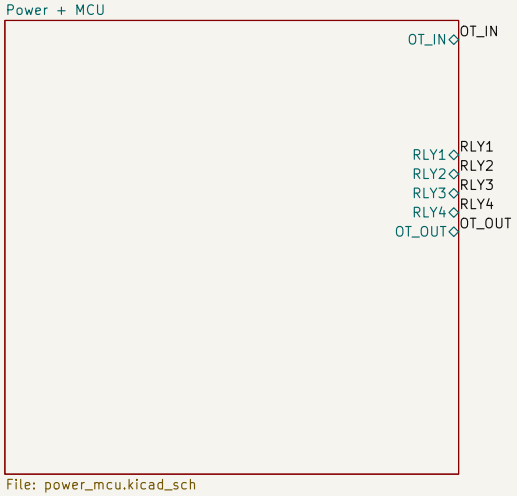
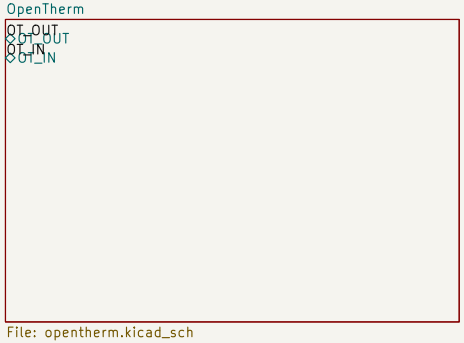
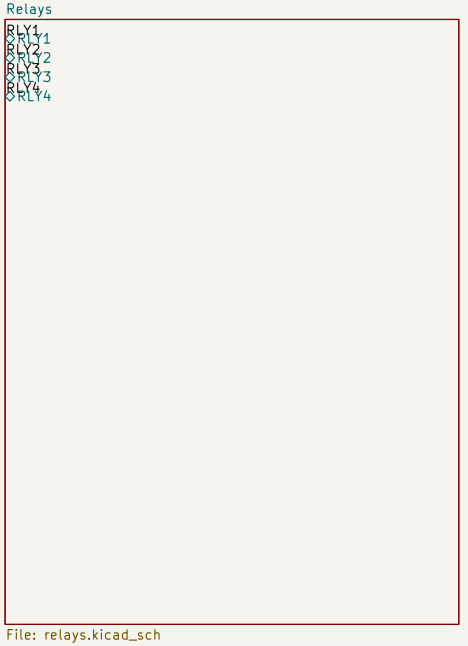






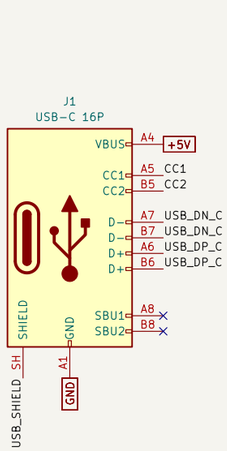
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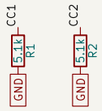
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File: heating-controller.kicad_sch

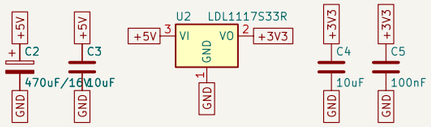
Title: Andiron

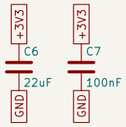
Size: A4	Date: 2026-10-22	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 1/4

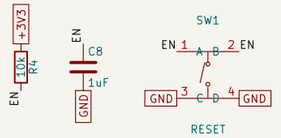
POWER + MCU. USB-C is the only supply and programmer (native USB-Serial-JTAG on IO12/IO13). Relay drives on IO21/22/23/3 only. Strapping: IO8 high (LED pull-up), IO9 BOOT, IO15 LED pull-up.

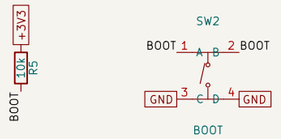


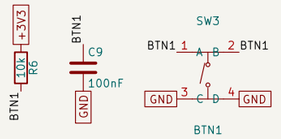


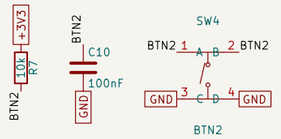


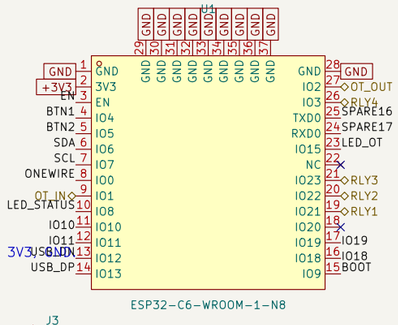


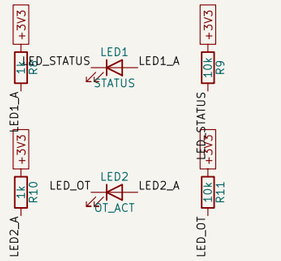


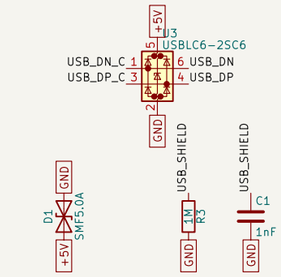












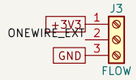
OLED header J5: GND, 3V3, SCL, SDA (standard 4-pin I2C OLED modules). Qwiic J6: GND, 3V3, SDA, SCL. Spare J7: IO16, IO17, IO10, IO11, IO18, IO19, 3V3, GND.

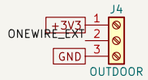


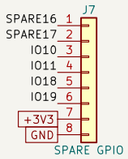


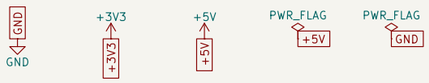










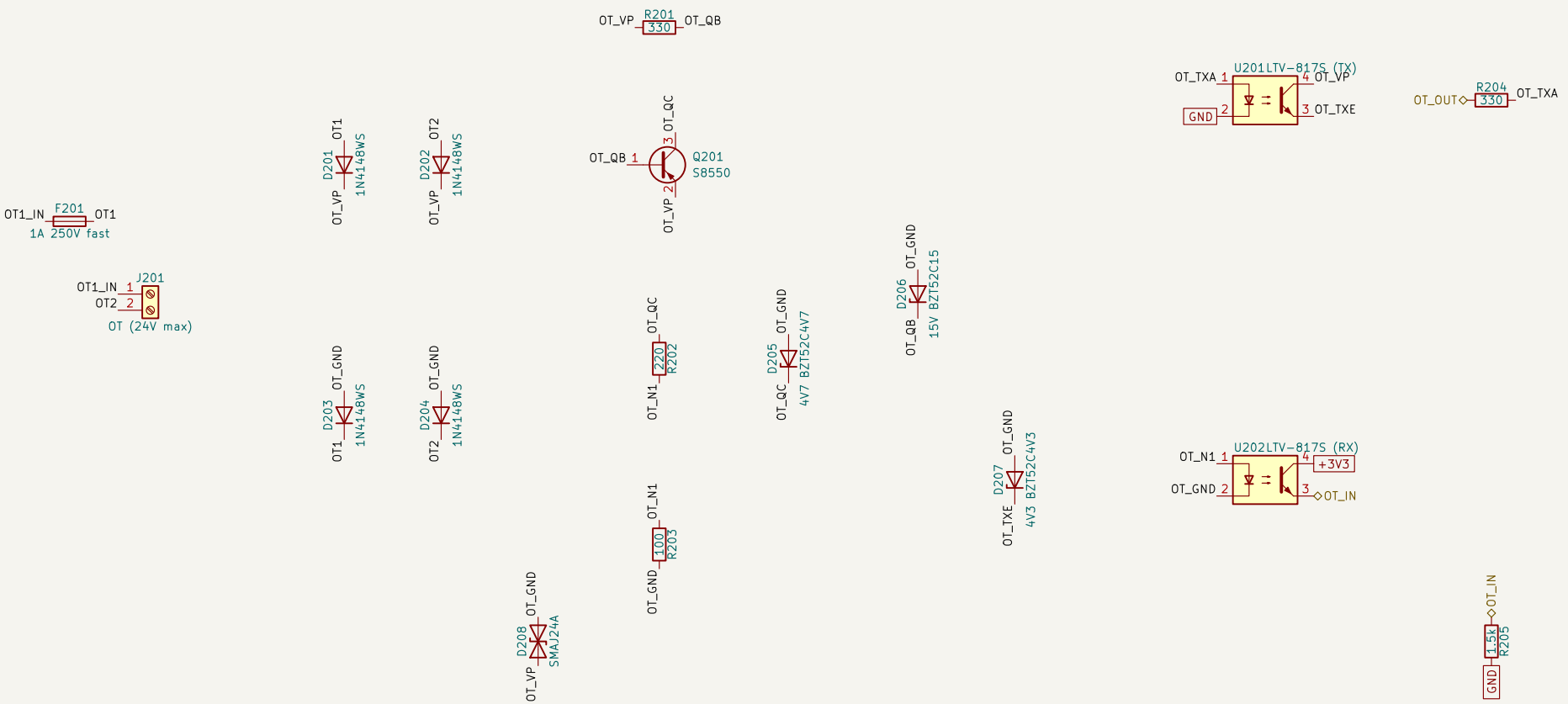


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Sheet: /Power + MCU/
File: power_mcu.kicad_sch

Title: Andiron

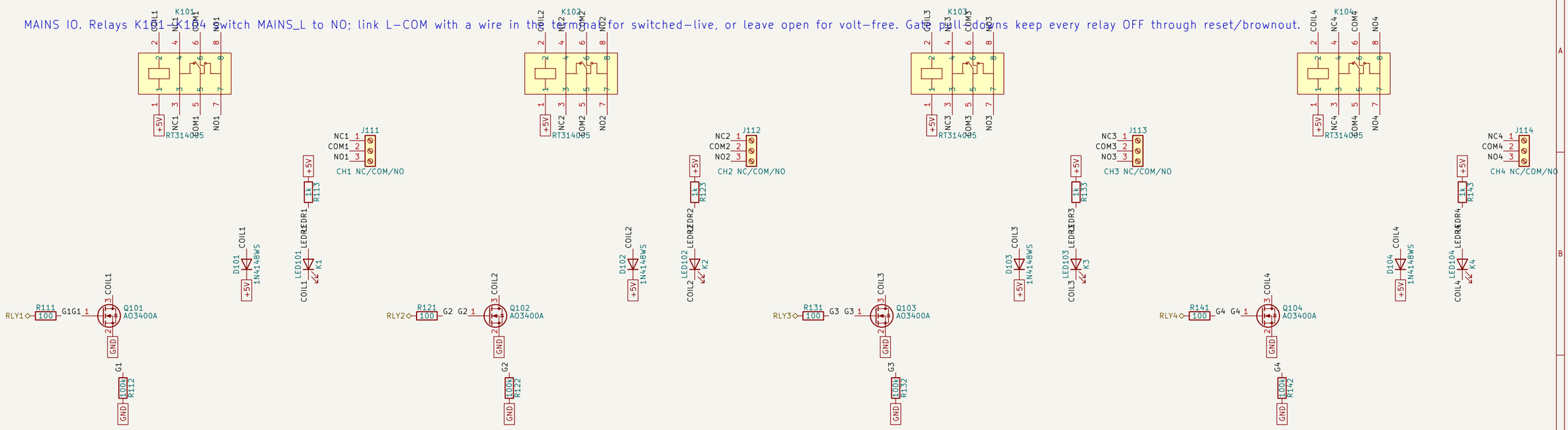
Size: A3	Date: 2026-10-22	**Rev: v1.0**
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OPENTHERM MASTER ADAPTER (Ihor Melnyk topology). Bus is boiler-powered SELV, polarity independent. OT_OUT high -> U201 on -> extra bus current (TX). Boiler voltage high -> D206 conducts



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Sheet: /OpenTherm/		
File: opentherm.kicad_sch		
Title: Andiron		
Size: A4	Date: 2026-10-22	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 3/4

MAINS IO. Relays K101–K104 switch MAINS_L to NO; link L–COM with a wire in the terminal for switched–live, or leave open for volt–free. Gate pull-downs keep every relay OFF through reset/brownout.



230V SENSE INPUTS: 6 x 22k 1206 chain (~1.7mA, ~420V impulse per part) -> LTV-817S LED with antiparallel diode. Output is ACTIVE-LOW when live present (inverted: true in ESPHome). ES1-3 return via MAINS_N. DEMAND: use 230V or 24V input with its own RET. MAINS_OK senses board L.

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Sheet: /Relays/
File: relays.kicad_sch

Title: Andiron

Size: A3	Date: 2026-10-22	**Rev: v1.0**
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